3.5 Smooth Manifolds — Problems

Work these alongside the schedule. Full solutions are in solutions.md. You may quote theorems from Lee, *Introduction to Smooth Manifolds* (2nd ed.) unless the problem asks you to prove them. Conventions: Lee's (determinant) convention for the wedge product; ∂M carries the Stokes (outward-normal-first) orientation.

Warm-up

1. Let $N = (0, \dots, 0, 1)$ and $S = -N$ in $S^n \subset \mathbb{R}^{n+1}$. Define stereographic projections $\sigma_N : S^n \setminus \{N\} \to \mathbb{R}^n$ and $\sigma_S : S^n \setminus \{S\} \to \mathbb{R}^n$ by

$$\sigma_N(x) = \frac{(x^1, \dots, x^n)}{1 - x^{n+1}}, \qquad \sigma_S(x) = \frac{(x^1, \dots, x^n)}{1 + x^{n+1}}.$$

Find σ_N^{-1}, compute the transition map $\sigma_S \circ \sigma_N^{-1}$, and conclude that $\{\sigma_N, \sigma_S\}$ is a smooth atlas.

2. Let $f : \mathbb{R}^3 \to \mathbb{R}$, $f(x, y, z) = x^2 + y^2 - z^2$.

(a) Find all regular values of f and conclude which level sets are embedded surfaces by the regular value theorem.
(b) Show that the cone $f^{-1}(0)$ is not a topological 2-manifold, hence not an embedded submanifold.

3. On $\mathbb{R}^3$ let $\omega = x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy$.

(a) Compute $d\omega$.
(b) Show that $\omega_p(u, v) = p \cdot (u \times v)$.
(c) Let $F(\varphi, \theta) = (\sin\varphi\cos\theta, \sin\varphi\sin\theta, \cos\varphi)$. Compute $F^*\omega$.

4. On $\mathbb{R}^2$ let $X = x\,\partial_y - y\,\partial_x$, $Y = \partial_x$, $Z = x\,\partial_x + y\,\partial_y$. Compute $[X, Y]$ and $[X, Z]$, and interpret the second answer in terms of flows.

5. On $M = \mathbb{R}^2 \setminus \{0\}$ let $\omega = \dfrac{x\,dy - y\,dx}{x^2 + y^2}$. Show that ω is closed but not exact. Show that its restriction to $\{x > 0\}$ is exact.

6. Show that $O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I\}$ is an embedded submanifold of $GL(n, \mathbb{R})$ of dimension $n(n-1)/2$, and identify $T_I O(n)$.

Core

7. Show that $\mathbb{RP}^n$ is orientable if and only if n is odd.

8. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and $\gamma : \mathbb{R} \to T^2 = S^1 \times S^1$, $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$. Show that γ is an injective immersion but not an embedding. (You may use that $\{n\alpha \bmod 1 : n \in \mathbb{Z}\}$ is dense in $[0, 1)$.)

9. (a) Find the maximal flow of $X = x^2\,\partial_x$ on $\mathbb{R}$, including its domain $\mathcal{D} \subseteq \mathbb{R} \times \mathbb{R}$, and show X is not complete.

(b) Find the flow of $Y = x\,\partial_x + 2y\,\partial_y$ on $\mathbb{R}^2$ and verify the group law $\theta_s \circ \theta_t = \theta_{s+t}$.

10. Let $X = x\,\partial_y - y\,\partial_x$ on $\mathbb{R}^2$ and $\omega = x\,dy$. Compute $\mathcal{L}_X\omega$ in two ways: (a) from the definition $\frac{d}{dt}\big|_{t=0} \theta_t^*\omega$ using the flow, and (b) by Cartan's formula.

11. Let ω be as in Problem 3 and give $S^2 = \partial\overline{B}^3$ the Stokes orientation. Compute $\int_{S^2}\omega$ directly using Problem 3(c), and again using Stokes' theorem.

12. Show that $H^1_{dR}(S^1) \cong \mathbb{R}$, with the isomorphism given by $[\omega] \mapsto \int_{S^1}\omega$.

13. Let M be a compact oriented smooth n-manifold without boundary, $n \geq 1$.

(a) Show $\int_M d\eta = 0$ for every $(n-1)$-form η.
(b) Deduce that no volume form on M is exact and that $H^n_{dR}(M) \neq 0$.
(c) Deduce that S^n is not diffeomorphic to $\mathbb{R}^n$ — and find a simpler reason too.

14. Let M be a nonempty compact smooth manifold and $k \geq 1$. Show that there is no smooth submersion $F : M \to \mathbb{R}^k$.

15. Prove that every smooth manifold admits a Riemannian metric.

16. (a) Show that T^2 is parallelizable (has a global frame of vector fields).

(b) Identify $\mathbb{R}^4$ with the quaternions $\mathbb{H}$ and S^3 with the unit quaternions. Show that $p \mapsto ip$, $p \mapsto jp$, $p \mapsto kp$ define a global orthonormal frame on S^3.
(c) Why does the same idea fail for S^2?

Qualifying-exam level

17. Compute $H^k_{dR}(S^n)$ for all k and $n \geq 1$ using the Mayer–Vietoris sequence. You may use the Poincaré lemma and homotopy invariance.

18. (a) Prove there is no smooth retraction $r : \overline{B}^n \to S^{n-1}$ ($n \geq 2$), using Stokes' theorem.

(b) Deduce that every smooth map $\overline{B}^n \to \overline{B}^n$ has a fixed point.

19. Let M be a connected, simply connected smooth manifold. Show that every closed 1-form on M is exact, i.e. $H^1_{dR}(M) = 0$.

20. On $\mathbb{R}^3$ let D be the rank-2 distribution spanned by $V = \partial_x + y\,\partial_z$ and $W = \partial_y$.

(a) Show $D = \ker(dz - y\,dx)$.
(b) Show D is not involutive, hence not integrable.
(c) Let $\alpha = dz - y\,dx$. Compute $\alpha \wedge d\alpha$ and explain how it detects (b).

21. Let X, Y be complete vector fields on M with flows θ, ψ. Prove that $[X, Y] = 0$ if and only if $\theta_t \circ \psi_s = \psi_s \circ \theta_t$ for all s, t. (You may use $\frac{d}{dt}\big|_{t=t_0} d(\theta_{-t})_{\theta_t(p)}(Y_{\theta_t(p)}) = d(\theta_{-t_0})([X,Y]_{\theta_{t_0}(p)})$.)

22. (a) Show that $d(\det)_A(B) = \det(A)\mathrm{tr}(A^{-1}B)$ for $A \in GL(n,\mathbb{R})$.

(b) Deduce that $SL(n,\mathbb{R})$ is an embedded submanifold of dimension $n^2 - 1$ with $T_I SL(n,\mathbb{R}) = \{B : \operatorname{tr} B = 0\}$.
(c) Show that $\det(e^B) = e^{\operatorname{tr} B}$ by considering the curve $t \mapsto \det(e^{tB})$, and conclude that exp maps trace-zero matrices into $SL(n,\mathbb{R})$.